

An IRT-Based Approach to Obtaining Item-Aware Learning Achievement

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Abstract

Due to the technical advances of computer networks, developing a learning assessment system with the diagnosis technology induced has become an important issue. As we know, data mining is a powerful tool in discovering hidden knowledge or rules such as Concept Effect Relationships from the testing results. However, since the learning achievement decided by the testing results may be affected by the profile of the test item, analyzing the learning achievement derived directly from the testing results without the preprocessing may cause the mining ineffective. Therefore, Item-Aware Learning Achievement, which considers the effect of the test item profile, is needed to solve this issue. In this paper, we first propose an IRT (Item Response Theory) Based Approach to obtain the Item-Aware Learning Achievement, and then apply the Item-Aware Learning Achievement in constructing the concept map to evaluate the performance of the IRT-Based Approach. The experimental results show that the Item-Aware Learning Achievement obtained by the proposed approach can improve the efficiency of the concept map construction.

1 Introduction

To discover the relationships among concepts, lots of researches have applied data mining approaches on the historical testing records in diagnosis [4][5][6]. However, they usually indicated the learning status of each test item by the binary score 0 or 1 to represent bad or good, respectively. And most of the students' learning achievement may be good if the test item is easy and may be bad if the test item is difficult. In other words, the contribution of the test item in deciding the learning achievement depends on the difficulty and the discrimination of the test item profile in addition to the examinee's learning ability. Therefore, the current learning achievement presented only by the correct/error answering ratio may be too rough to use in mining

analysis. It seems that the fuzzy measurement of the learning achievement with the consideration of the test item profile, which includes the difficulty and the discrimination is needed to obtain the Item-aware Learning Achievement. To solve this issue, we propose an Item Response Theory (IRT) Based Approach to derive the empirical relationships among examination data, which utilizes the test item profile and the examinee's learning ability to fuzzify the item's testing results.

However, the difficulty and the discrimination of the test item profile, which usually cannot be easily available, can be calculated by the correct answering ratio according to the Instruction Theory. We then implement the Test Item Analyzer to calculate the difficulty and the discrimination of test item profile, and utilize the Z-score, which is the relative standings of the examinee's score distribution, to present the learning ability of the student.

With the profile of test item and the student's learning ability, Fuzzy Learning Response Membership Function based upon the IRT is designed to fuzzify the learning achievement of each student corresponding to each test item. By according to the relative weight of the concepts and skills included in the test item, it is easy to derive the concept learning status from the test item. Thus we obtain the learning achievement of each concepts and skills included in the test item.

Furthermore, the learning achievement presented by the correct/error answering ratio treats the contribution of learning response in each test item as totally equal. If the test item have extreme difficulty, concept's learning achievement defined as the correct/error answering ratio of the test sheet is then affected dramatically, and causes the false judgment in students' learning. Sugeno Fuzzy Measure Function is well known aggregation function in fuzzy group decision and information fusion [7]. In order to rectify the disadvantages mentioned above and the consideration of the

concept hierarchy, Sugeno Fuzzy Measure Function is applied in our IRT-Based Approach to aggregate the WCR of the same concept derived from several test items, and finally obtains the Concept learning Response Aggregation index (CRA). The CRA is treated as the IALA of each concept and skill after an assessment, which includes the consideration of test item profile and the student's learning ability.

Finally, we verify the advantages of the IRT-Based Approach by applying the IALA to the Concept Map Construction [6] in high school Mathematics. The experimental results by applying IRT-Based Approach show more association rules can be generated, and circulated effect relationships can be reduced in the Concept Map Construction, which is better than the one without applying IRT-Based Approach.

2 Related Works

With the technology development of the assessment analysis, the effect relationships among concepts can be constructed by analyzing the testing results. For example, the Concept Effect Relationships (CER) [4][5] constructed by the CER Builder [3] provide the learning guidance of necessary concepts to enhance their learning performance. Based upon the historical learning records of students, the Two-Phase Fuzzy Mining and Learning Algorithm was proposed [8] to find the embedded association rules, which can be utilized to indicate the students' missing concepts during learning. However, the approaches mentioned above mine the raw data without the consideration of the difficulty and the discrimination of the test item may

confuse the relationships of the concepts.

Besides, Two-Phase Concept Map Construction (TP-CMC) algorithm [6] virtualizes the relationships among the concepts. But the learning achievement presented by the correct/error answering ratio may generate the noisy and circulated association rules, since the contribution of each test item is equal. Although deleting the testing result of low discrimination test item may solve the issue, some of the useful information can then be lost.

Therefore, to enhance the data analysis on the testing results, we propose an IRT-based Approach to obtain the IALA of the students with the consideration of the difficulty and the discrimination of the test item. Furthermore, with the consideration of the concept hierarchy in presenting the learning achievement, we applied Sugeno Fuzzy Measure Function to integrate the learning response of the same concept derived from different test items instead of the correct/error answering ratio.

3 IRT-Based Approach

The **IRT-Based Approach** shown in Figure 1 transforms the testing records of a test sheet into the Item-Aware Learning Achievement of the students. The Approach includes four procedures (Test Item Analyzer, IR Generator, Concept Response Computation and Concept Response Aggregation), and uses five tables (Testing Result Table (TRT), Item-Concept Relationship Table (ICRT), Student-Test Item Response Table (SIRT), Weight Concept Learning Response Mapping Table (CRMT), Item-Aware Learning Achievement Table (IALAT)).

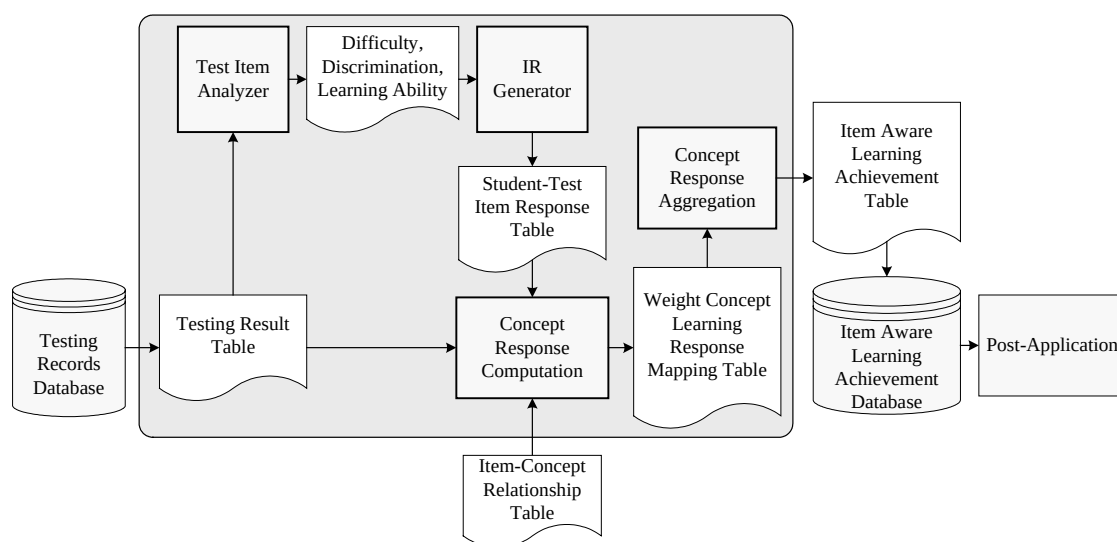


Figure 1. IRT-Based Approach

After inputting the raw data from the Testing Records Database, TRT is established by the student's answering result of each test item and is utilized by the Test Item Analyzer to calculate the difficulty, discrimination, score average and standard deviation of each test item, and also the Z-score of each student. With the difficulty, discrimination and Z-score, the Fuzzy Learning Response Membership Function of the IR Generator then calculates the IR of each student corresponding to each test item, which is stored in SIRT. Since ICRT represents the relative weight between the concept and the test item, WCR is calculated by the entries of SIRT, ICRT and TRT within the Concept Response Computation. Finally, Concept Response Aggregation applies the Sugeno Fuzzy Measure Function to aggregate the WCR of the same concept in several test items, and IALA of the students derived from the test sheet is then obtained.

3.1 Test Item Analyzer

Assume we have n students' testing records from the Testing Records Database, and the test sheet has m test items and p concepts. Let A be the matrix of TRT, which is denoted as

$$A = [a_{ji}]_{m \times n}, j=1,2,\dots,m, i=1,2,\dots,n, \quad (1)$$

where a_{ji} indicates the answered results of the j th test items from the i th student, whose value 1 and 0 denote the correct answered and wrong answered, respectively. Table 1 shows the example of TRT with four students tested by four items.

Test item Student ID	S_1	S_2	S_3	S_4
T_1	1	0	0	1
T_2	1	1	1	0
T_3	1	1	1	0
T_4	0	0	1	1

With the TRT, the score average and the standard deviation can be calculated by

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^m a_{ji}$$

and

$$S = \sqrt{\frac{\sum_{i=1}^n (\sum_{j=1}^m a_{ji} - \bar{X})^2}{n}},$$

respectively. Thus, we can obtain the Z-score as the i th student's ability by

$$\tilde{x}_i = \frac{x_i - \bar{X}}{S_X}, \quad i=1,2,\dots,n. \quad (2)$$

Suppose B and W are the set of high achievement students (the best 27%) and low achievement students (the worst 27%), respectively, let R_j^B and R_j^W be the correct answering ratio of the j th test item in the set B and the set W , respectively. Based upon the Instruction Theory, we define the Difficulty and the Discrimination of the j th test item as P_j and D_j :

$$P_j = \frac{R_j^B + R_j^W}{2}, \quad j=1,2,\dots,m \quad (3)$$

and

$$D_j = R_j^B - R_j^W, \quad j=1,2,\dots,m, \quad (4)$$

where difficulty value from 1 to 0 indicates the degree from easy to hard, and the discrimination value from 0 to 1 indicates the degree from low to high.

3.2 IR Generator

After having the difficulty P and the discrimination D , Two-Parameter Logistic Model of IRT can then be applied as the Fuzzy Learning Response Membership Function to obtain IR.

$$IR(D, P, \tilde{x}) = \frac{1}{1 + e^{-1.7D(\tilde{x}-P)}}, \quad (5)$$

where $\tilde{x}$ is the student's learning ability.

Suppose P_j and D_j are the difficulty and discrimination of the j th test item, respectively, and $\tilde{x}_i$ is the learning ability of the i th student. We denote the matrix of SIRT as

$$R = [r_{ji}]_{m \times n} = \left[\frac{1}{1 + e^{-1.7D_j(\tilde{x}_i - P_j)}} \right]_{m \times n}, \quad e = 2.719, \quad (6)$$

, where the entries indicate the i th student's learning status response from j th test item with value 0~1. Table 2 shows the example of SIRT with four students' IR corresponding to four test items.

Table 2 Student -test Item Response Table (SIRT)

Concept Test item	S_1	S_2	S_3	S_4
T_1	0.65	0.83	0.75	0.36
T_2	0.73	0.72	0.51	0.47
T_3	0.47	0.57	0.83	0.58
T_4	0.52	0.87	0.76	0.61

Example 1:

Suppose the student S_1 have learning ability $\tilde{x}=1.8$, and has correct answered the test item $\tilde{x}$ which difficulty $P=0.813$ and discrimination $D=0.375$. The student's IR responded to the test item is obtained by formula (5):

$$r_{11} = \frac{1}{1 + e^{-1.7 \times 0.375(1.8 - 0.813)}} = 0.65.$$

3.3 Concept Response Computation

Intuitively, a test item usually includes several concepts. Before we derive the concept learning status from the test item, the ICRT has to be firstly established. Suppose the test sheet has m test items including p concepts tested. We denote the matrix of ICRT as

$$F = (b_{jk})_{m \times p}, j=1,2,\dots,m, k=1,2,\dots,p, \quad (7)$$

where b_{jk} between 0 and 1 indicates the relativity weight of the k th concept that is tested by the j th test item. Table 3 shows the example of ICRT indicates the relationships between four items and four concepts.

Table 3 test Item-Concept Relation Table (ICRT)

Test item \ Concept	C_1	C_2	C_3	C_4
T_1	0.9	0.9	0.75	0
T_2	0.5	0	1	0
T_3	0	0	0.8	0
T_4	0.2	0.7	0	0.5

Finally, WCR w_{ijk} is obtained by multiplying the entries b_{jk} , a_{ji} and r_{ji} of the matrixes ICRT, TRT and SIRT, respectively. We define the matrix of CRMT as

$$W = [w_{ijk}]_{n \times m \times p} = [b_{jk} \cdot a_{ji} \cdot r_{ji}]_{n \times m \times p},$$

$$i=1,2,\dots,n, j=1,2,\dots,m, k=1,2,\dots,p, \quad (8)$$

where the value of w_{ijk} is between 0~1, which indicates the k th concept's learning status of the i th student derived from the j th test item's IR.

Example 2:

As shown in Table 1, the student S_1 has correctly answered the test item T_1 , and the IR is 0.65 according to the Table 2, and T_1 involves three concepts: C_1 , C_2 and C_3 shown in Table 3 with the relativity weight of 0.9, 0.9 and 0.75. Then the WCR derived from T_1 of S_1 concerning the concept C_1 , C_2 is

$$w_{111} = w_{112} = b_{11} \cdot a_{11} \cdot r_{11} = 0.9 \times 1 \times 0.65 = 0.585,$$

and the concept C_3 is

$$w_{113} = b_{13} \cdot a_{11} \cdot r_{11} = 0.75 \times 1 \times 0.65 = 0.488.$$

3.4 Concept Response Aggregation

We aggregate the WCR of the same concept in the test sheet in order to have an overall learning achievement of the concept. Given a universal set X , Since the Sugeno Fuzzy Measure Function is a set

function $g: 2^X \rightarrow [0,1]$ satisfies the following properties: (1) $g(\emptyset) = 0$, (2) $g(X) = 1$, and (3) if $A \subseteq B$, then $g(A) \subseteq g(B)$, where A and B are subsets of X , it seems to be appropriate as the aggregation function.

Let w_{ijk} be the entries of CRMT defined in function (9). Suppose the k th concept has a set of WCR $X_{ik} = \{w_{ijk} \mid j=1,2,3,\dots,m\}$ derived from m test items corresponding to the i th student. We apply the Sugeno Fuzzy Measure Function

$$g(X_{ik}) = \frac{1 - \prod_{j=1}^m (1 + \lambda \times w_{ijk})}{-\lambda}, \quad \lambda = -0.97 \quad (9)$$

to obtain the CRA, which is the IALA of the students derived from the test sheet. Figure 2 shows the conceptual diagram, where the CRAs of the three concepts C_1 , C_2 and C_3 are integrated by aggregating the WCR of the same concept derived from three test items T_1 , T_2 and T_3 .

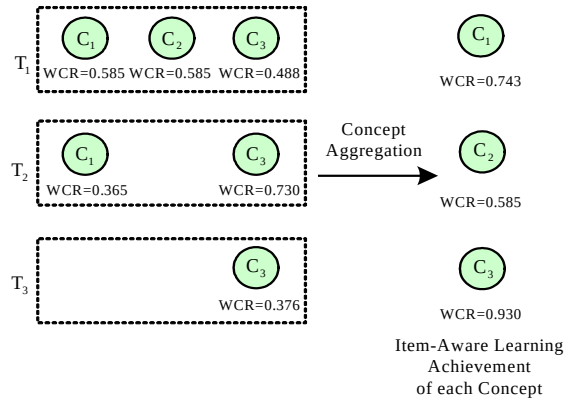


Figure 2 CRA index is aggregated from the same concept's WCR in each item's

We now define the matrix of IALAT as

$$H = [h_{ki}]_{p \times n} = \left[\frac{1 - \prod_{j=1}^m (1 - 0.97 \times w_{ijk})}{0.97} \right]_{p \times n},$$

$$i=1,2,\dots,n, k=1,2,\dots,p, \quad (10)$$

where h_{ki} is the k th Concept's Learning Achievement of the i th student with the value between 0~1, and w_{ijk} is the entry of CRMT defined in (9).

Example 3:

Suppose the student S_1 has the CRMT shown in Table 4. According to the entries of WCR by the columns, concept C_1 , C_2 and C_3 have the sets of WCR $\{w_{111}=0.585, w_{121}=0.585, w_{131}=0.488\}$, $\{w_{112}=0.325, w_{122}=0, w_{132}=0.650\}$ and $\{w_{113}=0, w_{123}=0, w_{133}=0.520\}$ derived from T_1 , T_2 and

T_3 , respectively.

Table 4 The CRMT with CRA

Item	C_1	C_2	C_3
T_1	0.585	0.585	0.488
T_2	0.325	0	0.650
T_3	0	0	0.520
CRA	0.726	0.585	0.931

Then the CRAs of the student S_1 corresponding to concepts C_1 , C_2 and C_3 are calculated as below.

$$h_{11} = \frac{1 - (1 - 0.97 \times 0.585)(1 - 0.97 \times 0.325)(1 - 0.97 \times 0.00)}{0.97} = 0.726$$

$$h_{21} = \frac{1 - (1 - 0.97 \times 0.585)(1 - 0.97 \times 0.00)(1 - 0.97 \times 0.00)}{0.97} = 0.585$$

$$h_{31} = \frac{1 - (1 - 0.97 \times 0.488)(1 - 0.97 \times 0.650)(1 - 0.97 \times 0.520)}{0.97} = 0.931$$

By the IRT-Based Approach, we transform the TRT of the testing records Database into the IALAT of the Item-Aware Concept Learning Achievement Database. Figure 1 illustrates the input and output of the IRT-Based Approach.

Testing results of Test item	Student			
	S_1	S_2	S_3	S_4
T_1	1	0	0	1
T_2	1	1	1	0
T_3	1	1	1	0
T_4	0	0	1	1
				
IALA	Student			
	S_1	S_2	S_3	S_4
C_1	0.684	0.877	0.803	0.550
C_2	0.520	0.915	0.859	0.617
C_3	0.826	0.962	0.947	0.803
C_4	0.000	0.435	0.380	0.305

Figure 3 TRT is transformed into the IALAT by IRT-Based Approach

4 Application

In this experiment, IALA is applied to the Concept Map Construction [6] to evaluate the advantages of the IRT-Based Approach. The basic data of Mathematics tests administered at a senior high school is given in Table 5. Where there are 42 students participated in the experiment, 32 items including 11 concepts with average test score = 62.36, and the average discrimination level of the test items = 0.107.

Table 5 Statistics of the experiment.

Course	Mathematics
School	Senior High School
Grade	K-11
Number of students	42
Average score	62.36
Number of test items	32
Number of concepts	11
Standard deviation of scores	15.43
Average difficulty of the test items	0.535
Difficulty range of the test items	0.125~0.938
Average discrimination of the test items	0.107
Discrimination range of the test items	-0.5~0.75

Table 6 lists the notation of concepts included in the test sheet. Besides, the weight of the concepts included in ICRT would be set to 1 to simplify our discussion. Moreover, the mining support is 0.6 and the confidence is 0.6.

Table 6 Notation of concepts included in the test sheet

Concept Notation	Concept
C_1	Spatial Relation of Points, Lines and Planes
C_2	Logical Concept
C_3	Symmetrical Point
C_4	Distance of Two Point
C_5	Trigonometric function
C_6	Cosine Theorem
C_7	Angle of Two Intersections Planes
C_8	Coordinates Reference of Spatial Object
C_9	Perpendicular Point
C_{10}	Equations of Coordinate Planes
C_{11}	Three Perpendicular Lines Theorem

In Figure 4 and Figure 5, the weight of the edge indicates the influent probability defined by the ordered pair, support and confidence, where support value is the probability of the event and the confidence is the conditional probability of the event based on the conditional event. For example in Figure 4(b), the link from the concept C_{10} to the concept C_3 has the ordered pair attribute value (0.95, 0.93), it represents the well learning probability of C_{10} is 0.95, based on such condition, the well learning probability of C_3 is 0.93. The attribute values of the node on Figure 4(a) and Figure 5(a) are IALA, and the attribute values of the node on Figure 4(b) and Figure 5(b) are the correct correct/error answering ratio of the concept. Before applying data mining, the attribute value of the

concept (node) is transformed from numeric to symbolic H/L at the same threshold 0.6. For example, let the transform threshold = 0.6, and the concept attribute value = 0.4 is transformed into symbolic L, since $0.4 < 0.6$. Figure 4 and Figure 5 show the H-H type CM and L-L type CM, respectively. Both types of the CM represent the assimilation effect relationships among concepts.

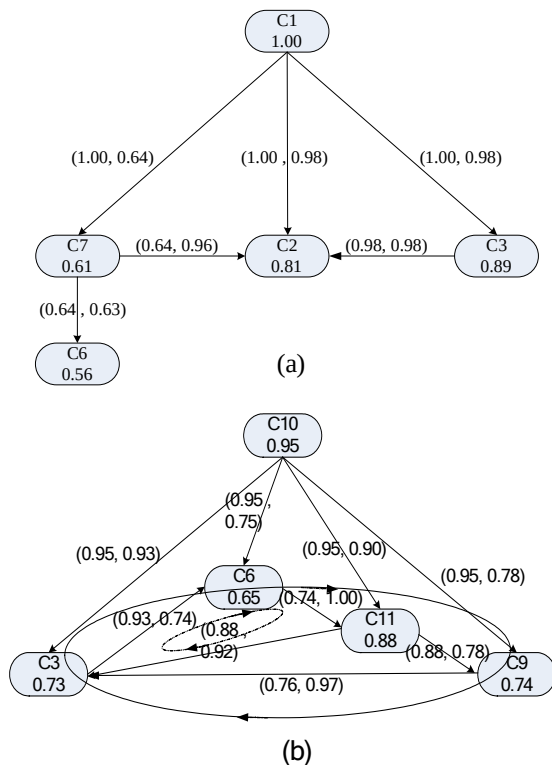


Figure 4 H-H type Concept Map constructed with and without applying IRT-Based Approach.

The notation $C_1 \rightarrow C_2$ in the H-H type CM represents the well learning of the prior concept C_1 implies the well learning of the concept C_2 . It makes sense that most of the students learn the prior concept easier than the further one.

The concept learning achievement represented by the correct answering ratio of the concept treats the contribution of learning response in each test item as totally equal, which does not consider the difficulty and the discrimination of the test item, thus the monotone association rules increase and the circulated concept relationships occurred. The CM without applying IRT-Based Approach is shown in Figure 4(b), which has two circulated concept relationships, one is the $C_3 \rightarrow C_6 \rightarrow C_{11} \rightarrow C_3$, and the other one is $C_3 \rightarrow C_6 \rightarrow C_{11} \rightarrow C_9 \rightarrow C_3$. As the experimental result shows, based on the same support and confidence, the CM construction with applying IRT-Based Approach reduces the circulated concept relationships, and is better than the one

without applying the IRT-Based Approach.

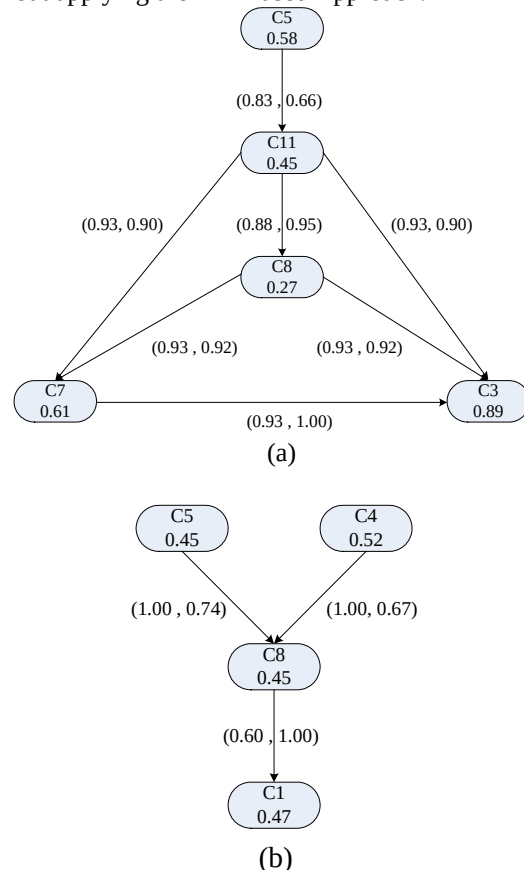


Figure 5 L-L type Concept Map constructed with and without applying IRT-Based Approach.

Figure 5 shows the L-L type Concept Map constructed with and without applying IRT-Based Approach. The notation $C_1 \rightarrow C_2$ in the L-L type CM represents the poor learning of the prior concept C_1 may cause the poor learning of the concept C_2 . For example, in Figure 5(a), if C_8 is not well learning by the students, the key problem in learning C_8 may be lack of understanding the concepts C_{11} with the conditional probability of 0.92, so the student should enhance the learning of concepts C_{11} instead of enhance the learning of concept C_8 . The CM of Figure 5(b) suggests the enhance learning of C_8 may have to enhance the learning of C_5 and C_4 with the conditional probability of 0.74 and 0.67, respectively.

5 Discussion

The learning achievement of the examinees decided by the test item depends on the difficulty and the discrimination of the test item in addition to the examinee's learning ability. As the results of the experiment, the concept C_{10} is tested by a lower

difficulty and discrimination test item. Based on the highest correct answering ratio, C_{10} is mined as the prior concept of the concept map in Figure 4(b). Meanwhile, with the consideration of the difficulty and the discrimination of the test item, the concept C_1 is mined as the prior concept of the concept map in Figure 4(a) based on the highest IALA obtained by the IRT-based Approach. Physically, according to Table 6, C_1 presents the concept of the Spatial Relation of Points, Lines and Planes and C_{10} presents the concept of Equations of Coordinate Planes, it seems to be more reasonable to treat concept C_1 as the prior concept instead of the concept C_{10} .

Furthermore, Concept learning status represented by the correct/error answering ratio of the concept, treats the contribution of learning achievement in each test item as totally equal. For example, the learning achievement of the students is all good and all bad if the test item is easy and difficult, respectively. Thus the mining based on such data will increase the number of the monotone association rules and the circulated association rules. Although the noisy and circulated association rules in constructing the concept map can be refined by deleting the low discrimination test item's testing score [6], some of the useful anomalous information may be somehow deleted too. Comparing the links based on the association rules of the CM in Figure 5, the links in Figure 5(b) is obviously less than the links in Figure 5(a) as the result mentioned above. The experiment shows IRT-Based Approach we propose is able to reduce the noisy and circulated association rules with all test item information reserved.

6 Conclusion

General speaking, the learning achievement derived directly from the testing results is affected by the profile of the test item and the examinee's learning ability, analyzing based on such learning achievement may cause the mining ineffective, such as noisy and circulated association rules. Consequently, we propose the Item-Aware Learning Achievement obtained by our IRT-Based Approach to rectify the testing results with the consideration of the difficulty and the discrimination of the test item. Besides, though the correct/error answering ratio is a convenient way to present the concept's learning achievement, it is also affected by the test item profile dramatically. Thus we apply the Sugeno Fuzzy Measure Function as the concept aggregation function based on rectified learning achievement to obtain Item-Aware Learning Achievement. By the way, with the Item-Aware Learning Achievement, Concept Map construction based on the

preprocessing of IRT-based Approach can have further applications such as relational graphic comparing and calculating, etc.

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